

Descriptive & Inferential Statistics for Surveys

58



Introduction to Descriptive Statistics for Survey Data

- Types of Survey Data:
 - Nominal,
 - Ordinal,
 - Interval,
 - Ratio

59

Descriptive vs Inferential Statistics

Descriptive Statistics	Inferential Statistics
Summarize data	Make predictions/inferences
What is happening	What might happen in the population
E.g., average satisfaction score = 4.2	E.g., Is that score significantly different from last year?

60



Population vs Sample

- **Population:**
 - All possible subjects.
- **Sample:**
 - Subset from the population.

61



Why Summarizing Data Matters

- Survey data can be **overwhelming**
 - Hundreds of rows
 - Dozens of variables
 - Different types
 - Without summarizing:
 - You can't see the **big picture**
 - Outliers go unnoticed
 - Trends stay hidden
 - **Descriptive statistics** help you:
 - Spot patterns
 - Compare groups
 - Clean and prepare the data for modeling
-

62



Example

- ***Average satisfaction score from 300 student responses is 4.2 what does that tell us?***
 - What kind of data is this?
 - Ordinal
 - What does 4.2 mean?
 - **Positive perception**
 - Is that enough?
 - Distribution
 - Next steps
-

63



Measures of Central Tendency and Dispersion

- **Central Tendency:**
 - Mean, Median, Mode
 - These are **summary measures** that help answer a simple question:
 - Where is the center of the data?
-

64



When to Use Which?

Measure	Best For
Mean	Interval/ratio data with no outliers
Median	Ordinal data or when outliers are present
Mode	Nominal/ordinal data or finding the most frequent category

65



Confidence Intervals (CI) for Survey Data

- A **confidence interval** provides a range within which we believe the **true population parameter** lies, based on a sample.
-

66



Why is this important in surveys?

- Surveys only collect data from a **sample**, not the entire population.
- So we need to express how **reliable** our estimates are.

Examples:

- “**62% of respondents support Policy X $\pm$ 4% (95% CI)**”
-

67



Why is this important in surveys?

- Surveys only collect data from a **sample**, not the entire population.
- So we need to express how **reliable** our estimates are.

Examples:

- “**62% of respondents support Policy X $\pm$ 4% (95% CI)**”
 - “**Average satisfaction score is 4.2 ± 0.3** ”
-

68



Margin of Error

- Margin of Error is the **maximum expected difference** between the **sample result** and the **true population value**.
-

69



Chi-Square Test

- Chi-Square Test for Independence is used when:
 - **Both variables are categorical.**
 - **Want to find relationship (association) between two categorical variables.**
 - **In survey data:**
 - *"Does gender influence whether someone would recommend a course?"*
 - *"Is age group related to satisfaction category?"*
-

70



Chi-Square Test

- To use Chi-square properly:
 - Observations must be independent.
 - Each individual (survey respondent) should only appear **once** in the dataset.
 - No participant should influence another's response.
-

71



Chi-Square Test

- To use Chi-square properly:
 - Observations must be independent.
 - Categories must be mutually exclusive.
 - Expected cell counts should generally be ≥ 5 for reliability.
-

72



Observed vs Expected Counts

- **Observed counts** (what you actually saw in the data)
 - **Expected counts** (what you would expect if there were **no association** between the variables)
-

73



Example

	Recommend Yes	Recommend No	Total
Male	120	80	200
Female	160	40	200
Total	280	120	400

74



Step 1: Set Up Hypotheses

- **Null Hypothesis (H_0):**
 - Gender and recommendation **are independent.**
- **Alternative Hypothesis (H_1):**
 - Gender and recommendation **are not independent.**

75

Step 2: Calculate Expected Counts

- Expected Count=(Row Total×Column Total)/Total
- For example:
 - Male, Recommend Yes: expected =(200*280)/400= 140
 - Male, Recommend No: expected =(200*120)/400= 60
 - Female, Recommend Yes: expected=(200*280)/400= 140
 - Female, Recommend No: expected =(200*120)/400= 60

	Recommend Yes	Recommend No	Row Total
Male	Observed: 120 / Expected: 140	Observed: 80 / Expected: 60	200
Female	Observed: 160 / Expected: 140	Observed: 40 / Expected: 60	200
Total	280	120	400

76

Step 3: Compute Chi-Square Statistic

- $\chi^2 = \sum \frac{(\text{Observed}-\text{Expected})^2}{\text{Expected}}$
- Male, Recommend Yes: (120-140)² / 140 =2.86
- Male, Recommend No: (80-60)² / 60 =6.67
- Female, Recommend Yes: (160-140)² / 140 =2.86
- Female, Recommend No: (40-60)² / 60 =6.67
- Total Chi-Square statistic: χ^2 =2.86+6.67+2.86+6.67=19.06

77